

Continuous X Activities in GigEase (CI / CD / CT / CM):-

The four Continuous X pillars of GigEase MLOps: Continuous Integration, Continuous Deployment, Continuous Training, and Continuous Monitoring, each keeping a different part of the system always up to date.

Continuous X	Activity	What It Involves	GigEase Example (from the platform docs)
Continuous Integration (CI)	Automated code & pipeline testing on every commit	Linting, unit tests, and integration tests run automatically on each pull request	CI runs against every change to the income, fraud, and risk pipeline code before merge
	Data & schema validation checks	New code is tested against expected data contracts before merging	CI blocks a merge that would break the 2-of-4 source RSMD confirmation logic (NewsAPI, NDMA, Maps, Police)
	Automated build & containerization	Every merge automatically builds and versions a container image	Each of the 4 agents (Work History, KYC, Fraud, Decision) is rebuilt and tagged automatically on merge
	Artifact & dependency versioning	Model and library versions are locked and tracked per build	LightGBM, scikit-learn (Isolation Forest), and SHAP versions are pinned to prevent explanation drift between builds
Continuous Deployment (CD)	Automated staged rollout	Passing builds move through canary or shadow deployment automatically	New risk-model versions automatically shadow-score live traffic before replacing the fortnightly-increment model
	Champion/challenger promotion	A challenger only replaces the champion if it wins on the defined metric	A challenger fraud model is auto-promoted only if it beats the champion on the cost-weighted objective
	Zero-downtime release strategy	Traffic is swapped between environments without interrupting service	Blue-green deployment keeps claim payouts processing uninterrupted during a fraud-model version swap
	Automated rollback on failure	A bad release reverts to the last known-good version without manual steps	A version breaching AUC/RMSE-drift thresholds rolls back automatically to the previous production model
	Scheduled retraining per model cadence	Training pipelines fire automatically on a fixed schedule	Income model retrains weekly, risk model fortnightly, and fraud model quarterly through pipeline jobs

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Continuous Training (CT)	Drift-triggered retraining	Retraining also fires off live performance signals, not just the calendar	A retrain triggers automatically when $AUC < 0.75$ or $RMSE\text{-}drift > 15\%$ is detected
	Automated data-validation gates	A training run is rejected if input data is missing or stale	Blocks a retrain if weather data is over 15 minutes stale or GPS pings are missing
	Feedback-loop enrichment	Real-world outcomes and appeals feed back into the next training set	Appeal outcomes from the 30-day IRDAI escalation path enrich the labeled dataset for the next fraud-model retrain
Continuous Monitoring (CM)	Real-time performance dashboards	AUC, RMSE-drift, and calibration are tracked live in production	Dashboards surface $AUC < 0.75$ and $RMSE\text{-}drift > 15\%$ the moment either threshold is crossed
	Data & concept-drift monitoring	Live data distribution is compared against training-time distribution	Checks whether Chennai-monsoon-trained thresholds still hold against Delhi heatwave conditions
	Data-freshness & ingestion monitoring	Alerts fire when a feed falls behind its expected polling cadence	Alerts if the 15-minute weather poll or 60-second GPS poll falls behind schedule
	Cross-agent traceability & audit logging	Every stage's output is logged so a bad decision can be traced end to end	A bad payout is traced through Work History -> KYC -> Fraud -> Decision agent logs to isolate the cause